Chapter 3. Graphical Causal Models

In Chapter 1 you saw how causal inference can be broken down into two problems: identification and estimation. In this chapter, you'll dive deeper into the identification part, which is arguably the most challenging one. This chapter is mostly theoretical, as you will be playing with graphical models without necessarily estimating their parameters with data. Don't let this fool you. Identification is the heart of causal inference, so learning its theory is fundamental for tackling causal problems in real life. In this chapter, you will:

- Get an introduction to graphical models, where you will learn what a graphical model for causality is, how associations flow in a graph, and how to query a graph using off-the-shelf software.
- Revisit the concept of identification through the lens of graphical models.
- Learn about two very common sources of bias that hinder identification, their causal graph structure, and what you can do about them.

Thinking About Causality

Have you ever noticed how those cooks in YouTube videos are excellent at describing food? "Reduce the sauce until it reaches a velvety consistency." If you are just learning to cook, you have no idea what this even means. Just give me the time I should leave this thing on the stove, will you! With causality, it's the same thing. Suppose you walk into a bar and hear folks discussing causality (probably a bar next to an economics department). In that case, you will hear them say how the confounding of income made it challenging to identify the effect of immigration on that neighborhood unemployment rate, so they had to use an instrumental variable. And by now, you might not understand what they are talking about. That's OK. You've only scratched the surface when it comes to understanding the language of causal inference. You've learned a bit about counterfactual outcomes and biases; enough so you could understand the key issue causal inference is trying to solve. Enough to appreciate what's going on behind the most powerful tool of causal inference: randomized controlled trials. But this tool won't always be available or simply won't work (as you'll soon see in "Selection Bias"). As you encounter more challenging causal inference problems, you'll also need a broader understanding of the causal inference language, so you can properly understand what you are facing and how to deal with it.

A well-articulated language allows you to think clearly. This chapter is about broadening your causal inference vocabulary. You can think of graphical models as one of the fundamental languages of causality. They are a powerful way of structuring a causal inference problem and making identification assumptions explicit, or even visual. Graphical models will allow you to make your thoughts transparent to others and to yourself.

STRUCTURAL CAUSAL MODEL

Some scientists use the term *structural causal model* (SCM) to refer to a unifying language of causal inference. These models are composed of graphs and causal equations. Here, I'll mostly focus on the graph aspect of SCMs.

As a starting point into the fantastic world of graphs, let's take our previous example of estimating the impact of emails on conversion. In that example, the treatment T is cross-sell email and the outcome Y is if a customer converted to a new product or not:

```
In [1]: import pandas as pd
        import numpy as np

        data = pd.read_csv("./data/cross_sell_email.csv")
        data
```

	gender	cross_sell_email	age	con
0	0	short	15	0
1	1	short	27	0
2	1	long	17	0
...	...	...	...	...
320	0	no_email	15	0
321	1	no_email	16	0
322	1	long	24	1

Let's also recall from the previous chapter that, in this problem, T is randomized. Hence, you can say that the treatment is independent from the potential outcomes, $(Y_0, Y_1) \perp T$, which makes association equal to causation:

$$E[Y_1 - Y_0] = E[Y|T = 1] - E[Y|T = 0]$$

Importantly, there is absolutely no way of telling that the independence assumption holds just by looking at the data. You can only say that it does because you have information about the treatment assignment mechanism. That is, you know that emails were randomized.

Visualizing Causal Relationships

You can encode this knowledge in a graph, which captures your beliefs about what causes what. In this simple example, let's say you believe that cross-sell emails cause conversion. You also believe that the other variables you measured, age and gender, also cause conversion. Moreover, you can also add variables you didn't measure to the graph. We usually denote them by the letter U, since they are unobserved. There are probably many unobserved variables that cause conversion (like customer income, social background), and age (how your product appeals to different demographics, the city the company is operating in). But since you don't measure them, you can bundle everything into a U node that represents all those unmeasured variables. Finally, you can add a randomization node pointing to T, representing your knowledge of the fact that the cross-sell email was randomized.

DAG

You might find people referring to causal graphs as *DAGs*. The acronym stands for directed acyclic graph. The directed part tells you that the edges have a direction, as opposed to undirected graphs, like a social network, for example. The acyclic part tells you that the graph has no loops or cycles. Causal graphs are usually directed and acyclic because causality is nonreversible.

To add those beliefs of yours to a graph and literally see them, you can use `graphviz`:

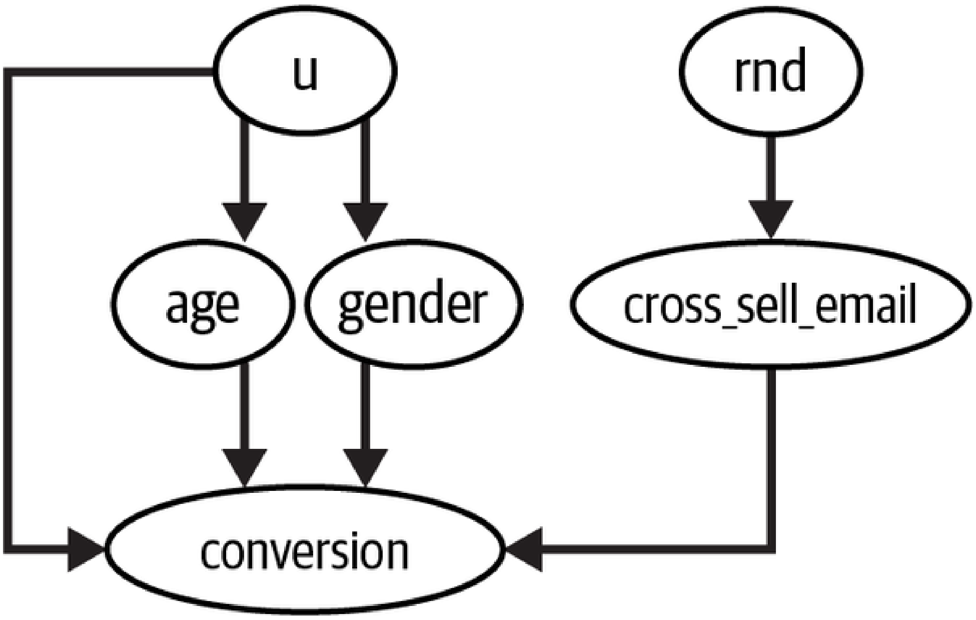
```
In [2]: import graphviz as gr

        g_cross_sell = gr.Digraph()

        g_cross_sell.edge("U", "conversion")
        g_cross_sell.edge("U", "age")
        g_cross_sell.edge("U", "gender")

        g_cross_sell.edge("rnd", "cross_sell_email")
        g_cross_sell.edge("cross_sell_email", "conversion")
        g_cross_sell.edge("age", "conversion")
        g_cross_sell.edge("gender", "conversion")

        g_cross_sell
```



Each node in the graph is a random variable. You can use arrows, or edges, to show if a variable causes another. In this graphical model, you are saying that email causes conversion, that U causes age, conversion, and gender, and so on and so forth. This language of graphical models will help you clarify your thinking about causality, as it makes your beliefs about how the world works explicit. If you are pondering how impractical this is—after all, there is no way you are going to encode all the hundreds of variables that are commonly present in today's data applications—rest assured you won't need to. In practice, you can radically simplify things, by bundling up nodes, while also keeping the general causal story you are trying to convey. For example, you can take the preceding graph and bundle the observable variables into an X node. Since they both are caused by U and cause conversion, your causal story remains intact by joining them.

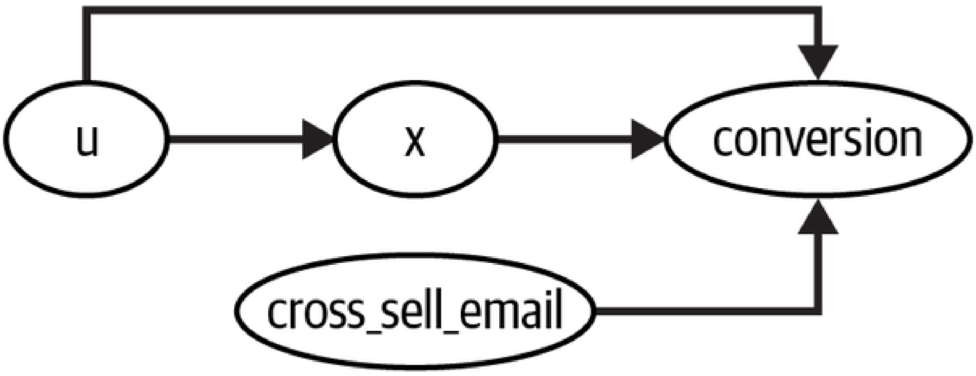
Also, when you are representing variables that have been randomized or intervened on, you can just remove all incoming arrows from it:

```
In [3]: # rankdir:LR layers the graph from left to right
        g_cross_sell = gr.Digraph(graph_attr={"rankdir": "I

        g_cross_sell.edge("U", "conversion")
        g_cross_sell.edge("U", "X")
```

```
g_cross_sell.edge("cross_sell_email", "conversion")
g_cross_sell.edge("X", "conversion")

g_cross_sell
```

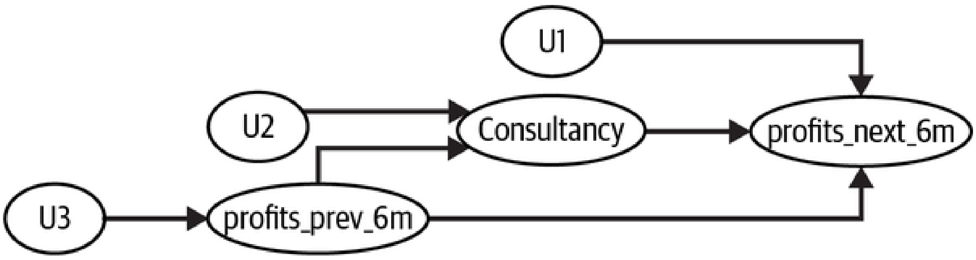


What is interesting to realize here is that perhaps the most important information in a DAG is actually what is *not* in it: an edge missing from one variable to another means there is an assumption of no direct causal link between the two. For example, in the preceding graph, you are assuming that nothing causes both the treatment and the outcome.

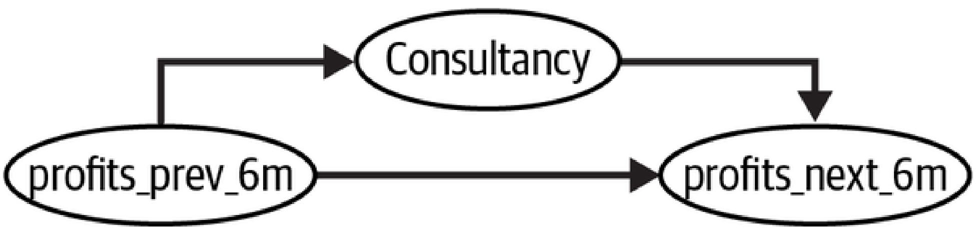
Just like with every language you learn, you are probably looking into this and thinking it doesn't all make complete sense. That's normal. I could just throw at you a bunch of rules and best practices to represent causal relationships between variables in a graph. But that is probably the least efficient way of learning. Instead, my plan is to simply expose you to lots and lots of examples. With time, you will get the hang of it. For now, I just want you to keep in mind that graphs are a very powerful tool for understanding why association isn't causation.

Are Consultants Worth It?

To see the power of DAGs, let's consider a more interesting example, where the treatment is not randomized. Let's suppose you are the manager of a company contemplating the decision of whether to bring in some top-notch consultants. You know that they are expensive, but you also know that they have expert knowledge from working with the best companies in the business. To make things more complicated, you are not sure if the top-notch consultants will improve your business or if it is just the case that only very profitable businesses can afford those consultants, which is why their presence correlates with strong business performance. It would be awesome if someone had randomized the presence of consultants, as this would make answering the question trivial. But of course you don't have that luxury, so you will have to come up with something else. As you can probably see by now, this is a problem of untangling causation from association. To understand it, you can encode your beliefs about its causal mechanisms in a graph:



Notice how I've added U nodes to each of these variables to represent the fact that there are other things we can't measure causing them. Since graphs usually represent random variables, it is expected that a random component will cause all the variables, which is what those Us represent. However, they won't add anything to the causal story I'm going to tell, so I might just as well omit them:



Here, I'm saying that the past performance of a company causes the company to hire a top-notch consultant. If the company is doing great, it can afford to pay the expensive service. If the company is not doing so great, it can't. Hence, past performance (measured here by past profits) is what determines the odds of a company hiring a consultant. Remember that this relationship is not necessarily deterministic. I'm just saying that companies that are doing well are *more likely* to hire top-notch consultants.

Not only that, companies that did well in the past 6 months are very likely to also do well in the next 6 months. Of course, this doesn't always happen, but, on average, it does, which is why you also have an edge from past performance to future performance. Finally, I've added an edge from consultancy to the firm's future performance. Your goal is to know the strengths of this connection. This is the causal relationship you care about. Does consultancy actually cause company performance to increase?

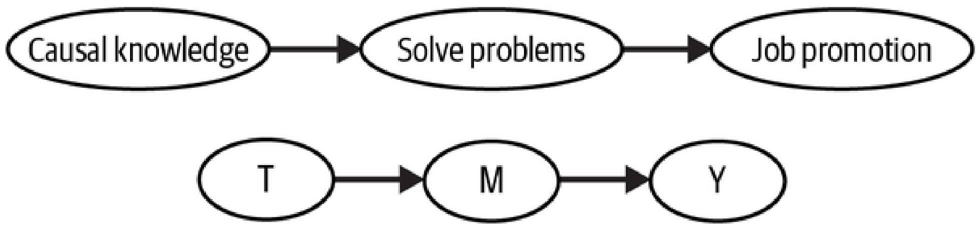
Answering this question is not straightforward because there are two sources of association between consultancy and future performance. One is causal and the other is not. To understand and untangle them, you first need to take a quick look at how association flows in causal graphs.

Crash Course in Graphical Models

Schools offer whole semesters on graphical models. By all means, if you want to go deep in graphical models, it will be very beneficial for your understanding of causal inference. But, for the purpose of this book, it is just (utterly) important that *you understand what kind of independence and conditional independence assumptions a graphical model entails*. As you'll see, associations flow through a graphical model as water flows through a stream. You can stop this flow or enable it, depending on how you treat the variables in the graph. To understand this, let's examine some common graphical structures and examples. They will be pretty straightforward, but they are the sufficient building blocks to understand everything about the flow of association, independence, and conditional independence on graphical models.

Chains

First, look at this very simple graph. It's called a chain. Here T causes M, which causes Y. You can sometimes refer to the intermediary node as a mediator, because it mediates the relationship between T and Y:



In this first graph, although causation only flows in the direction of the arrows, association flows both ways. To give a more concrete example, let's say that knowing about causal inference improves your problem-solving skills, and problem solving increases your chances of getting a promotion. So causal knowledge causes your problem-solving skills to increase, which in turn causes you to get a job promotion. You can say here that job promotion is dependent on causal knowledge. The greater the causal expertise, the greater your chances of getting a promotion. Also, the greater your chances of promotion, the greater your chance of having causal knowledge. Otherwise, it would be difficult to get a promotion. In other words, job promotion is associated with causal inference expertise the same way that causal inference expertise is associated with job promotion, even though only one of the directions is causal. When two variables are associated with each other, you can say they are dependent or not independent:

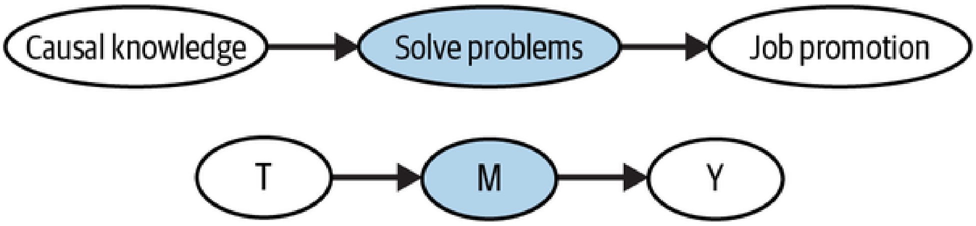
$$T \not\perp Y$$

Now, let's hold the intermediary variable fixed. You could do that by looking only at people with the same M, or problem-solving skills in our example.

Formally, you can say you are conditioning on M. In this case, the dependence is blocked. So, T and Y are independent given M. You can write this mathematically as:

$$T \perp Y | M$$

To indicate that we are conditioning on a node, I'll shade it:



To see what this means in our example, think about conditioning on people's problem-solving skills. If you look at a bunch of people with the same problem-solving skills, knowing which ones are good at causal inference doesn't give any further information about their chances of getting a job promotion. In mathematical terms:

$$E[Promotion|Solve\ problems, Causal\ knowledge] = E[Promotion|Solve\ problems]$$

The inverse is also true; once I know how good you are at solving problems, knowing about your job promotion status gives me no further information about how likely you are to know causal inference.

As a general rule, if you have a chain like in the preceding graph, association flowing in the path from T to Y is blocked when you condition on an intermediary variable M. Or:

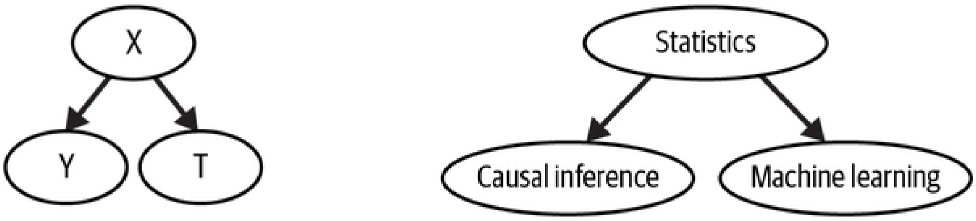
$$T \not\perp Y$$

but

$$T \perp Y | M$$

Forks

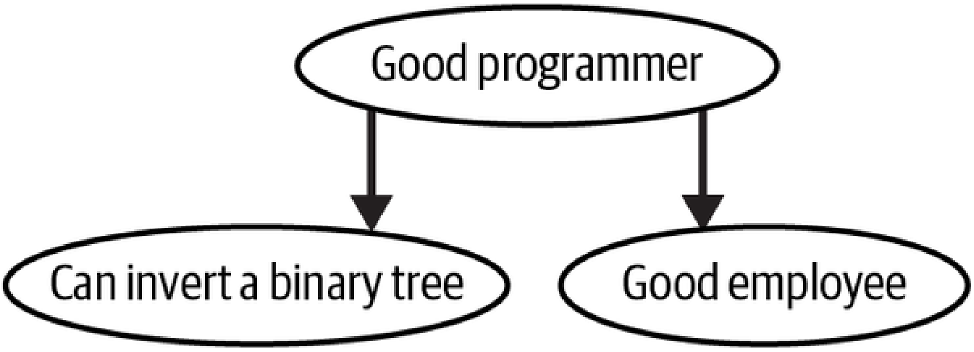
Moving on, let's consider a fork structure. In this structure, you have a common cause: the same variable causes two other variables down the graph. In forks, association flows backward through the arrows:



For example, let's say your knowledge of statistics causes you to know more about causal inference and about machine learning. However, knowing causal inference doesn't help you with machine learning and vice versa, so there is no edge between those variables.

This graph is telling you that if you don't know someone's level of statistical knowledge, then knowing that they are good at causal inference makes it more likely that they are also good at machine learning, even if causal inference doesn't help you with machine learning. That is because even if you don't know someone's level of statistical knowledge, you can infer it from their causal inference knowledge. If they are good at causal inference, they are probably good at statistics, making it more likely that they are also good at machine learning. The variables at the tip of a fork move together even if they don't cause each other, simply because they are both caused by the same thing. In the causal inference literature, when we have a common cause between a treatment and the outcome, we call that common cause a *confounder*.

The fork structure is so important in causal inference that it deserves another example. Do you know how tech recruiters sometimes ask you to solve problems that you'll probably never find in the job you are applying for? Like when they ask you to invert a binary tree or count duplicate elements in Python? Well, they are essentially leveraging the fact that association flows through a fork structure in the following graph:



The recruiter knows that good programmers tend to be top performers. But when they interview you, they don't know if you are a good programmer or not. So they ask you a question that you'll only be able to answer if you are. That question doesn't have to be about a problem that you'll encounter in the job you are applying for. It just signals whether you are a good programmer or not. If you can answer the question, you will likely be a good programmer, which means you will also likely be a good employee.

Now, let's say that the recruiter already knows that you are a good programmer. Maybe they know you from previous companies or you have an impressive degree. In this case, knowing whether or not you can answer the application process questions gives no further information on whether you will be a good employee or not. In technical terms, you can say that answering the question and being a good employee are independent, once you condition on being a good programmer.

More generally, if you have a fork structure, two variables that share a common cause are dependent, but independent when you condition on the common cause. Or:

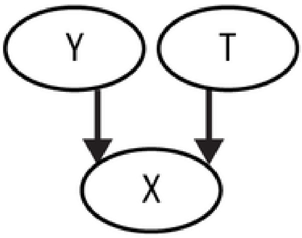
$T \not\perp Y$

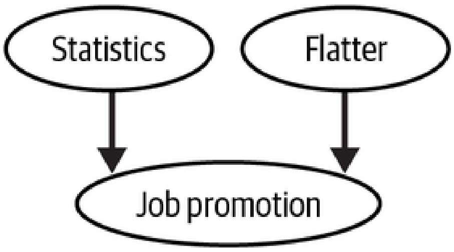
but

$T \perp Y | X$

Immorality or Collider

The only structure missing is the immorality (and yes, this is a technical term). An *immorality* is when two nodes share a child, but there is no direct relationship between them. Another way of saying this is that two variables share a common effect. This common effect is often referred to as a *collider*, since two arrows collide at it:





In an immorality, the two parent nodes are independent of each other. But they become dependent if you condition on the common effect. For example, consider that there are two ways to get a job promotion. You can either be good at statistics or flatter your boss. If I don't condition on your job promotion, that is, I don't know if you will or won't get it, then your level of statistics and flattering are independent. In other words, knowing how good you are at statistics tells me nothing about how good you are at flattering your boss. On the other hand, if you did get a job promotion, suddenly, knowing your level of statistics tells me about your flattering level. If you are bad at statistics and did get a promotion, you will likely be good at flattering your boss. Otherwise, it will be very unlikely for you to get a promotion. Conversely, if you are good at statistics, it is more likely that you are bad at flattering, as being good at statistics already explains your promotion. This phenomenon is sometimes called *explaining away*, because one cause already explains the effect, making the other cause less likely.

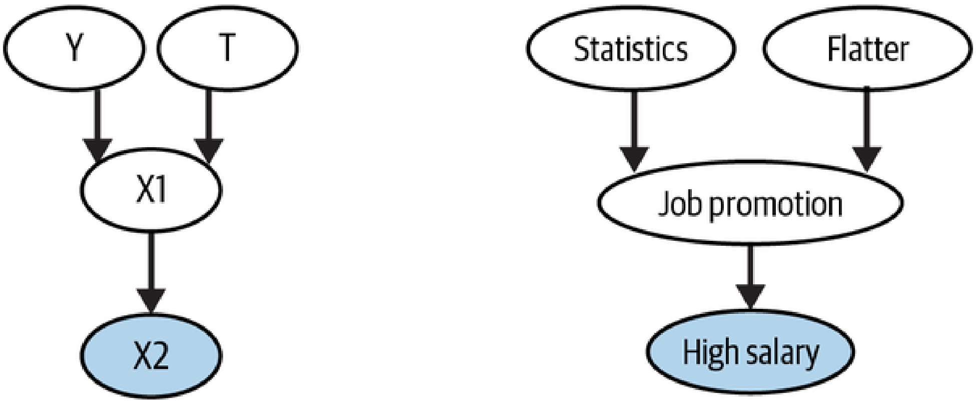
As a general rule, conditioning on a collider opens the association path, making the variables dependent. Not conditioning on it leaves it closed. Or:

$T \perp Y$

and

$T \not\perp Y|X$

Importantly, you can open the same dependence path if instead of conditioning on the collider, you condition on a effect (direct or not) of the collider. Continuing with our example, let's now say that getting a job promotion massively increases your salary, which gives you the next graph:



In this graph, even if you don't condition on the collider, but condition on a cause of it, the causes of the collider become dependent. For instance, even if I don't know about your promotion, but I do know about your massive salary, your knowledge about statistics and boss flattering become dependent: having one makes it less likely that you also have the other.

The Flow of Association Cheat Sheet

Knowing these three structures—chains, forks, and immoralities—you can derive an even more general rule about independence and the flow of association in a graph.

A path is blocked if and only if:

1. It contains a non-collider that has been conditioned on.
2. It contains a collider that has not been conditioned on and has no descendants that have been conditioned on.

Figure 3-1 is a cheat sheet about how dependence flows in a graph.

If these rules seem a bit opaque or hard to grasp, now is a good time for me to tell you that, thankfully, you can use off-the-shelf algorithms to check if two variables in a graph are associated with each other or if they are independent. To tie everything you learned together, let's go over a final example so I can show you how to code it up.

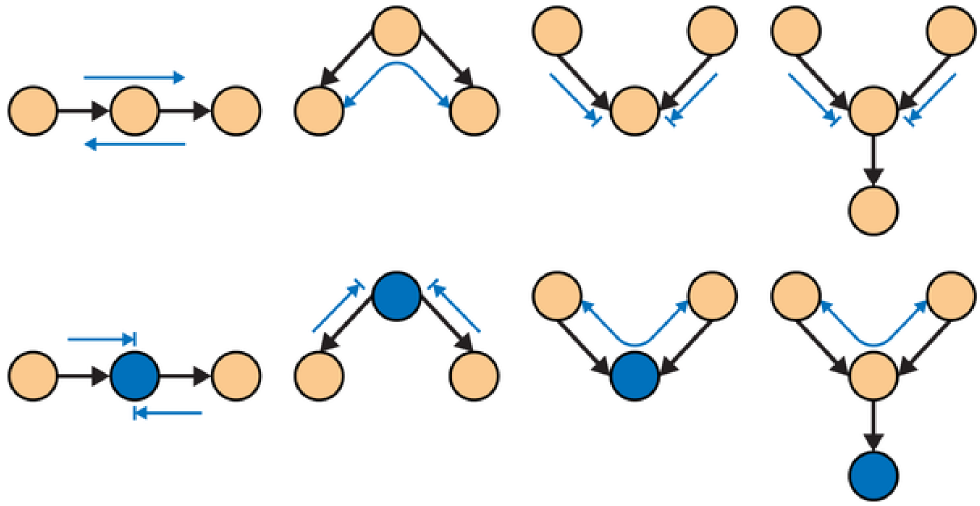
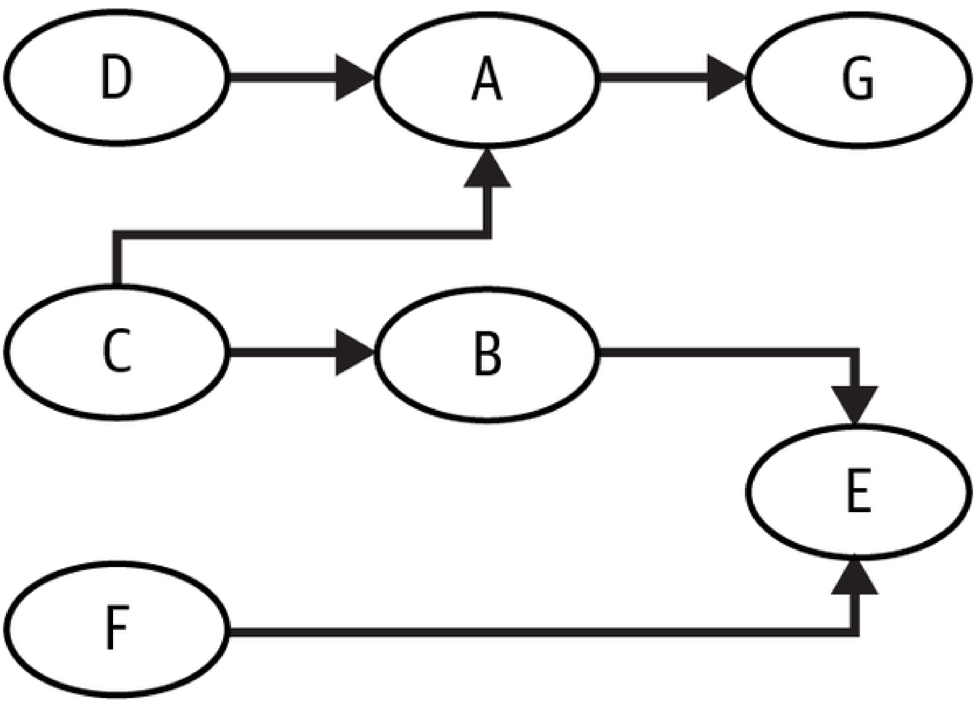


Figure 3-1. Cheat sheet about how dependence flows in a graph

Querying a Graph in Python

Take the following graph:



In a moment, you'll input this graph to a Python library that will make answering questions about it pretty easy. But before you do that, as an exercise to internalize the concepts you've just learned, try to answer the following questions on your own:

Are D and C dependent?

Are D and C dependent given A?

Are D and C dependent given G?

Are A and B dependent?

Are A and B dependent given C?

Are G and F dependent?

Are G and F dependent given E?

Now, to see if you got them right, you can input that graph into a `DiGraph`, from `networkx`. `networkx` is a library to handle graphical models and has a bunch of handy algorithms that will help you inspect this graph:

```
In [4]: import networkx as nx

        model = nx.DiGraph([
            ("C", "A"),
            ("C", "B"),
            ("D", "A"),
            ("B", "E"),
            ("F", "E"),
            ("A", "G"),
        ])
```

As a starter, let's take D and C. They form the immorality structure you saw earlier, with A being a collider. From the rule about independence in an immorality structure, you know that D and C are independent. You also know that if you condition on the collider A, association starts to flow between them. The method `d_separated` tells you if association flows between two variables in the graph (d-separation is another way of expressing the independence between two variables in a graph). To condition on a variable, you can add it to the observed set. For example, to check if D and C are dependent given A, you can use `d_separated` and pass the fourth argument `z={"A"}` :

```
In [5]: print("Are D and C dependent?")
        print(not(nx.d_separated(model, {"D"}, {"C"}, {})))

        print("Are D and C dependent given A?")
        print(not(nx.d_separated(model, {"D"}, {"C"}, {"A"

        print("Are D and C dependent given G?")
        print(not(nx.d_separated(model, {"D"}, {"C"}, {"G"
```

```
Out[5]: Are D and C dependent?
        False
        Are D and C dependent given A?
        True
        Are D and C dependent given G?
        True
```

Next, notice that D, A, and G form a chain. You know that association flows in a chain, so D is not independent from G. However, if you condition on the intermediary variable A, you block the flow of association:

```
In [6]: print("Are G and D dependent?")
        print(not(nx.d_separated(model, {"G"}, {"D"}, {})))

        print("Are G and D dependent given A?")
        print(not(nx.d_separated(model, {"G"}, {"D"}, {"A"
```

```
Out[6]: Are G and D dependent?
        True
        Are G and D dependent given A?
        False
```

The last structure you need to review is the fork. You can see that A, B, and C form a fork, with C being a common cause of A and B. You know that association flows through a fork, so A and B are not independent. However, if you condition on the common cause, the path of association is blocked:

```
In [7]: print("Are A and B dependent?")
        print(not(nx.d_separated(model, {"A"}, {"B"}, {})))

        print("Are A and B dependent given C?")
        print(not(nx.d_separated(model, {"A"}, {"B"}, {"C"}
```

```
Out[7]: Are A and B dependent?
        True
        Are A and B dependent given C?
        False
```

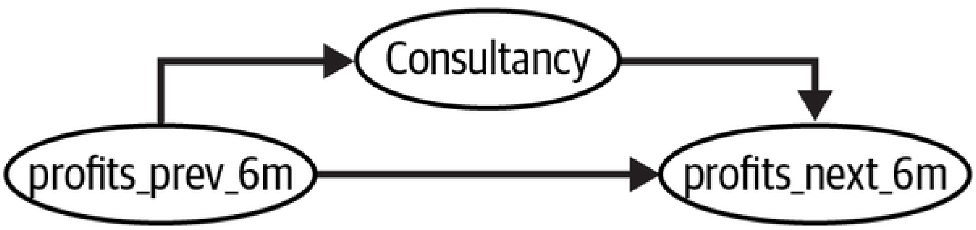
Finally, let's put everything together and talk about G and F. Does association flow between them? Let's start at G. You know that association flows between G and E, since they are in a fork. However, association stops at the collider E, which means that G and F are independent. Yet if you condition on E, association starts to flow through the collider and the path opens, connecting G and F:

```
In [8]: print("Are G and F dependent?")
        print(not(nx.d_separated(model, {"G"}, {"F"}, {})))

        print("Are G and F dependent given E?")
        print(not(nx.d_separated(model, {"G"}, {"F"}, {"E"]
```

```
Out[8]: Are G and F dependent?
        False
        Are G and F dependent given E?
        True
```

This is great. Not only did you learn the three basics structures in graphs, you also saw how to use off-the-shelf algorithms to check for independences in the graph. But what does this have to do with causal inference? It's time to go back to the problem we were exploring at the beginning of the chapter. Recall that we were trying to understand the impact of hiring expensive, top-notch consultants on business performance, which we depicted as the following graph:



You can use your newly acquired skills to see why association is not causation in this graph. Notice that you have a fork structure in this graph. Therefore, there are two flows of association between consultancy and company's future performance: a direct causal path and a noncausal path that is confounded by a common cause. This latter one is referred to as a *backdoor path*. The presence of a confounding backdoor path in this graph demonstrates that the observed association between consultancy and company performance cannot be solely attributed to a causal relationship.

Understanding how associations flow in a graph through noncausal paths will allow you to be much more precise when talking about the difference between association and causation. For this reason, it pays to revisit the concept of identification, now under the new light of graphical models.

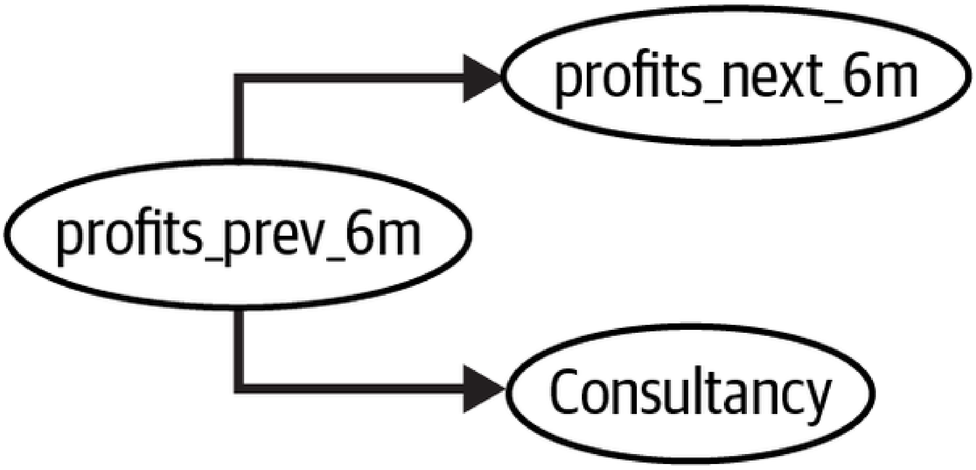
Identification Revisited

So far, in the absence of randomization, the argument I've been using to explain why it is so hard to find the causal effect is that treated and untreated are not comparable to each other. For example, companies that hire consultants usually have better past performance than those that don't hire expensive consultants. This results in the sort of bias you've seen before:

$$E[Y|T=1] - E[Y|T=0] = \underbrace{E[Y_1 - Y_0|T=1]}_{ATT} + \underbrace{\{E[Y_0|T=1] - E[Y_0|T=0]\}}_{BIAS}$$

Now that you've learned about causal graphs, you can be more precise about the nature of that bias and, more importantly, you can understand what you can do to make it go away. Identification is intimately related to independence in a graphical model. If you have a graph that depicts the causal relationship between the treatment, the outcome, and other relevant variables, you can think about identification as the process of isolating the causal relationship between the treatment and the outcome in that graph. During the identification phase, you will essentially close all undesirable flows of association.

Take the consultancy graph. As you saw earlier, there are two association paths between the treatment and the outcome, but only one of them is causal. You can check for bias by creating a causal graph that is just like the original one, but with the causal relationship removed. If treatment and outcome are still connected in this graph, it must be due to a noncausal path, which indicates the presence of bias:

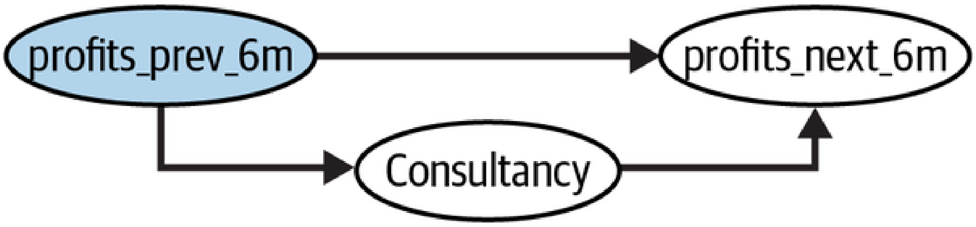



```
In [9]: consultancy_model_severed = nx.DiGraph([
            ("profits_prev_6m", "profits_next_6m"),
            ("profits_prev_6m", "consultancy"),
        #      ("consultancy", "profits_next_6m"), # causal
        ])

        not(nx.d_separated(consultancy_model_severed,
                           {"consultancy"}, {"profits_next_
```

```
Out[9]: True
```

These noncausal flows of association are referred to as *backdoor paths*. To identify the causal relationship between T and Y, you need to close them so that only the causal path one remains. In the consultancy example, you know that conditioning on the common cause, the company's past performance, closes that path:



CIA and the Adjustment Formula

You just saw that conditioning on `profits_prev_6m` blocks the noncausal association flow between the treatment, consultancy, and the outcome—the company's future performance. As a result, if you look at a group of companies with similar past performance and, inside that group, compare the future performance of those that hired consultants with those that didn't, the difference can be entirely attributed to the consultants. This makes intuitive sense, right? The difference in future performance between the treated (companies that hired consultants) and the untreated is 1) due to the treatment itself and 2) due to the fact that companies that hire consultants tend to be doing well to begin with. If you just compare treated and untreated companies that are doing equally well, the second source of difference disappears.

Of course, like with everything in causal inference, you are making an assumption here. Specifically, you are assuming that all sources of noncausal association between the treated and the outcome is due to the common causes you can measure and condition on. This is very much like the independence assumption you saw earlier, but in its weaker form:

$$(Y_0, Y_1) \perp T | X$$

This *conditional independence assumption* (CIA) states that, if you compare units (i.e.,companies) with the same level of covariates X, their potential outcomes will be, on average, the same. Another way of saying this is that treatment seems *as if it were randomized*, if you look at units with the same values of covariate X.

CIA NAMES

The CIA permeates a lot of causal inference research and it goes by many names, like ignorability, exogeneity, or exchangeability.

The CIA also motivates a very simple way to identify the causal effect from observable quantities in the data. If treatment looks as good as random within groups of X, all you need to do is compare treated and untreated inside each of the X defined groups and average the result using the size of the group as weights:

$$ATE = E_X[E[Y|T=1] - E[Y|T=0]]$$

$$\begin{aligned} ATE &= \sum_x \{(E[Y|T=1, X=x] - E[Y|T=0, X=x])P(X=x)\} \\ &= \sum_x \{E[Y|T=1, X=x]P(X=x) - E[Y|T=0, X=x]P(X=x)\} \end{aligned}$$

This is called the *adjustment formula* or conditionality principle. It says that, if you condition on or control for X, the average treatment effect can be identified as the weighted average of in-group differences between treated and control. Again, if conditioning on X blocks the flow of association through the noncausal paths in the graph, a causal quantity, like the ATE, becomes identifiable, meaning that you can compute it from observable data. The process of closing backdoor paths by adjusting for confounders gets the incredibly creative name of *backdoor adjustment*.

Positivity Assumption

The adjustment formula also highlights the importance of positivity. Since you are averaging the difference between treatment and outcome over X, you must ensure that, for all groups of X, there are some units in the treatment and some in the control, otherwise the difference is undefined. More formally, you can say that the conditional probability of the treatment needs to be strictly positive and below 1: $1 > P(T|X) > 0$. Identification is still possible when positivity is violated, but it will require you to make dangerous extrapolations.

POSITIVITY NAMES

Since the positivity assumption is also very popular in causal inference, it too goes by many names, like common support or overlap.

An Identification Example with Data

Since this might be getting a bit abstract, let's see how it all plays out with some data. To keep our example, let's say you've collected data on six companies, three of which had low profits (1 million USD) in the past six months and three of which had high profits. Just like you suspected, highly profitable companies are more likely to hire consultants. Two out of the three high-profit companies hired them, while only one out of the three low-profit companies hired consultants (if the low sample bothers you, please pretend that each data point here actually represents 10,000 companies):

```
In [10]: df = pd.DataFrame(dict(
             profits_prev_6m=[1.0, 1.0, 1.0, 5.0, 5.0, 5.0]
             consultancy=[0, 0, 1, 0, 1, 1],
             profits_next_6m=[1, 1.1, 1.2, 5.5, 5.7, 5.7],
         ))

         df
```

	profits_prev_6m	consultancy	profits_next_6m
0	1.0	0	1.0
1	1.0	0	1.1
2	1.0	1	1.2
3	5.0	0	5.5
4	5.0	1	5.7
5	5.0	1	5.7

If you simply compare `profits_next_6m` of the companies that hired consultants with those that didn't, you'll get a difference of 1.66 MM in profits:

```
In [11]: (df.query("consultancy==1")["profits_next_6m"].mea
          - df.query("consultancy==0")["profits_next_6m"].r
```

```
Out[11]: 1.666666666666667
```

But you know better. This is not the causal effect of consultancy on a company's performance, since the companies that performed better in the past are overrepresented in the group that hired consultants. To get an unbiased estimate of the effect of consultants, you need to look at companies with similar past performance. As you can see, this yields more modest results:

```
In [12]: avg_df = (df
                   .groupby(["consultancy", "profits_prev_6
                   ["profits_next_6m"]
                   .mean())

         avg_df.loc[1] - avg_df.loc[0]
```

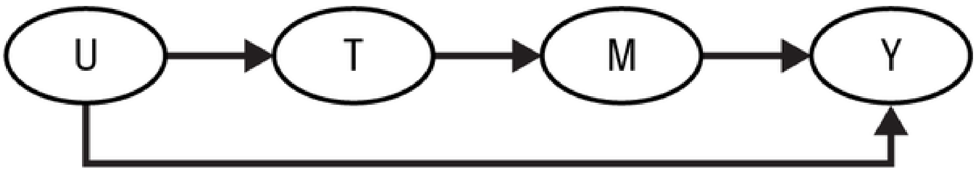
```
Out[12]: profits_prev_6m
         1.0     0.15
         5.0     0.20
         Name: profits_next_6m, dtype: float64
```

If you take the weighted average of these effects, where the weights are the size of each group, you end up with an unbiased estimate of the ATE. Here, since the two groups are of equal size, this is just a simple average, giving you an ATE of 175,000. Hence, if you are a manager deciding whether to hire consultants and you are presented with the preceding data, you can conclude that the impact of consultants on future profits is about 175k USD. Of course, in order to do that, you have to invoke the CIA. That is, you have to assume that past performance is the only common cause of hiring consultants and future performance.

You just went through a whole example of encoding your beliefs about a causal mechanism into a graph and using that graph to find out which variables you needed to condition on in order to estimate the ATE, even without randomizing the treatment. Then, you saw what that looked like with some data, where you estimated the ATE, following the adjustment formula and assuming conditional independence. The tools used here are fairly general and will inform you of many causal problems to come. Still, I don't think we are done yet. Some graphical structures—and the bias they entail—are much more common than others. It is worth going through them so you can start to get a feeling for the difficulties that lie ahead in your causal inference journey.

FRONT DOOR ADJUSTMENT

The backdoor adjustment is not the only possible strategy to identify causal effects. One can leverage the knowledge of causal mechanisms to identify the causal effect via a front door, even in the presence of unmeasured common causes:

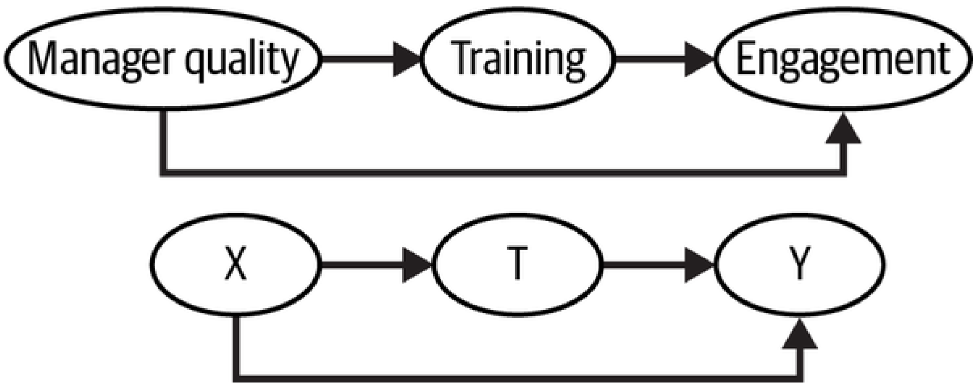


With this strategy, you must be able to identify the effect of the treatment on a mediator and the effect of that mediator on the outcome. Then, the identification of the effect of treatment on the outcome becomes the combination of those two effects. However, in the tech industry, it's hard to find applications where such a graph is plausible, which is why the front door adjustment is not so popular.

Confounding Bias

The first significant cause of bias is confounding. It's the bias we've been discussing so far. Now, we are just putting a name to it. *Confounding happens when there is an open backdoor path through which association flows noncausally, usually because the treatment and the outcome share a common cause*. For example, let's say that you work in HR and you want to know if your new management training program is increasing employers' engagement. However, since the training is optional, you believe only managers that are already doing great attend the program and those who need it the most, don't. When you measure engagement of the teams under the managers that took the training, it is much higher than that of the teams under the managers who didn't attend the training. But it's hard to know how much of this is causal. Since there is a common cause between treatment and outcome, they would move together regardless of a causal effect.

To identify that causal effect, you need to close all backdoor paths between the treatment and the outcome. If you do so, the only effect that will be left is the direct effect $T \rightarrow Y$. In our example, you could somehow control for the manager's quality prior to taking the training. In that situation, the difference in the outcome will be only due to the training, since manager quality prior to the training would be held constant between treatment and control. Simply put, *to adjust for confounding bias, you need to adjust for the common causes of the treatment and the outcome*:

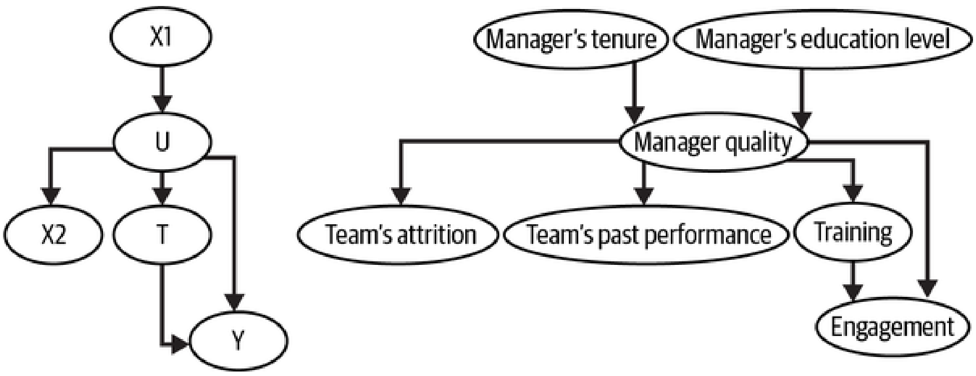


Unfortunately, it is not always possible to adjust for all common causes. Sometimes, there are unknown causes or known causes that you can't measure. The case of manager quality is one of them. Despite all the effort, we still haven't yet figured out how to measure management quality. If you can't observe manager quality, then you can't condition on it and the effect of training on engagement is not identifiable.

Surrogate Confounding

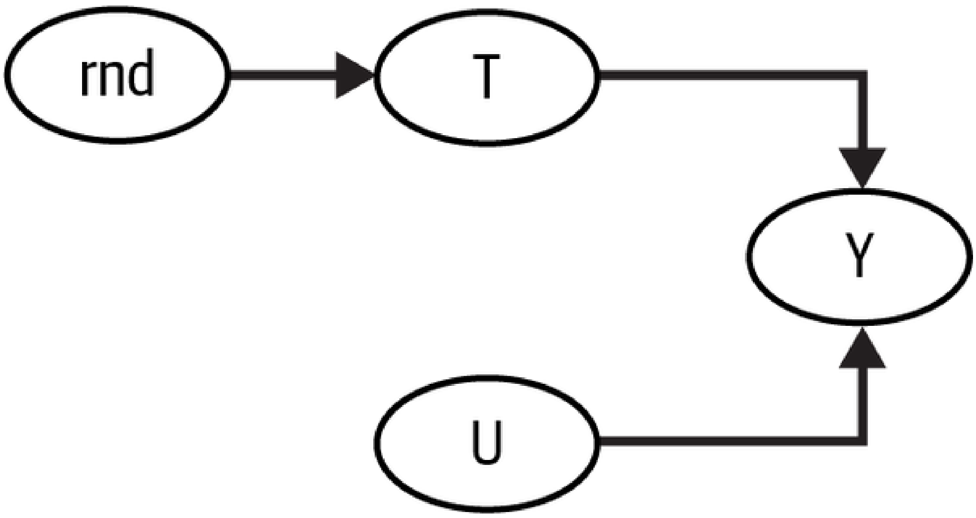
In some situations, you can't close all the backdoor paths due to unmeasured confounders. In the following example, once again, manager quality causes managers to opt in for the training and team's engagement. So there is confounding in the relationship between the treatment (training) and the outcome (team's engagement). But in this case, you can't condition on the confounder, because it is unmeasurable. In this case, the causal effect of the treatment on the outcome is not identifiable due to confounder bias. However, you have other measured variables that can act as proxies for the confounder manager's quality. Those variables are not in the backdoor path, but controlling for them will help reduce the bias (even though it won't eliminate it). Those variables are sometimes referred to as surrogate confounders.

In this example, you can't measure manager quality, but you can measure some of its causes, like the manager's tenure or level of education; and some of its effects, like the team's attrition or performance. Controlling for those surrogate variables is not sufficient to eliminate bias, but it sure helps:



Randomization Revisited

In many important and very relevant research questions, confounders are a major issue, since you can never be sure that you've controlled for all of them. But if you are planning to use causal inference mostly in the industry, I have good news for you. In the industry, you are mostly interested in learning the causal effects of things that your company can control—like prices, customer service, and marketing budget—so that you can optimize them. In those situations, it is fairly easy to know what the confounders are, because the business usually knows what information it used to allocate the treatment. Not only that, even when it doesn't, it's almost always an option to intervene on the treatment variable. This is precisely the point of A/B tests. When you randomize the treatment, you can go from a graph with unobservable confounders to one where the only cause of the treatment is randomness:



Consequently, besides trying to see what variables you need to condition on in order to identify the effect, you should also be asking yourself what are the possible interventions you could make that would change the graph into one where the causal quantity of interest is identifiable.

Not all is lost when you have unobserved confounders. In Part IV, I'll cover methods that can leverage time structure in the data to deal with unobserved confounders. Part V will cover the use of instrumental variables for the same purpose.

SENSITIVITY ANALYSIS AND PARTIAL IDENTIFICATION

When you can't measure all common causes, instead of simply giving up, it is often much more fruitful to shift the discussion from "Am I measuring all confounders?" to "How strong should the unmeasured confounders be to change my analysis significantly?" This is the main idea behind sensitivity analysis. For a comprehensible review on this topic, I suggest you check out the paper "Making Sense of Sensitivity: Extending Omitted Variable Bias," by Cinelli and Hazlett.

Additionally, even when the causal quantity you care about can't be point identified, you can still use observable data to place bounds around it. This process is called *partial identification* and it is an active area of research.

Selection Bias

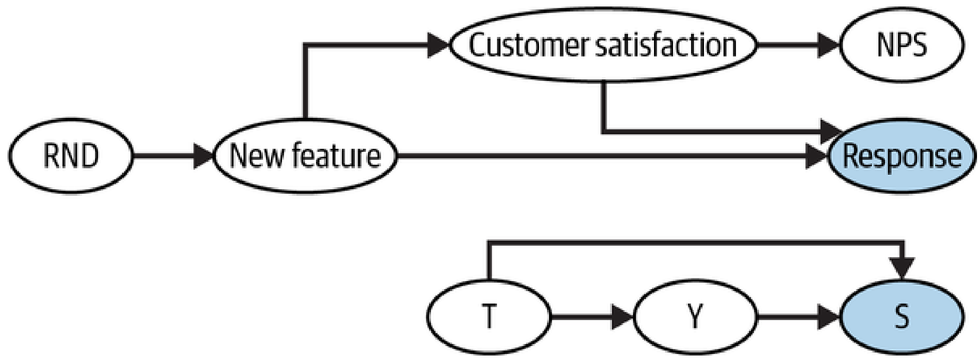
If you think confounding bias was already a sneaky little stone in your causal inference shoe, just wait until you hear about selection bias. While confounding bias happens when you don't control for common causes to the treatment and outcome, selection bias is more related to conditioning on common effects and mediators.

BIAS TERMINOLOGY

There isn't a consensus in the literature on the names of biases. For instance, economists tend to refer to all sorts of biases as selection bias. In contrast, some scientists like to further segment what I'm calling selection bias into collider bias and mediator bias. I'll use the same terminology as in the book *Causal Inference: What If*, by Miguel A. Hernán and James M. Robins (Chapman & Hall/CRC).

Conditioning on a Collider

Consider the case where you work for a software company and want to estimate the impact of a new feature you've just implemented. To avoid any sort of confounding bias, you do a randomized rollout of the feature: 10% of the customers are randomly chosen to get the new feature, while the rest don't. You want to know if this feature made your customer happier and more satisfied. Since satisfaction isn't directly measurable, you use Net Promoter Score (NPS) as a proxy for it. To measure NPS, you send a survey to the customers in the rollout (treated) and in the control groups, asking them if they would recommend your product. When the results arrive, you see that the customers who had the new feature and responded to the NPS survey had higher NPS scores than the ones that didn't have the new feature and also responded to the NPS survey. Can you say that this difference is entirely due to the causal effect of the new feature on NPS? To answer this question, you should start with the graph that represents this situation:



To cut to the chase, sadly, the answer is no. The issue here is that you can only measure NPS for those who responded to the NPS survey. You are estimating the difference between treated and control while also conditioning on customers who responded to the NPS survey. Even though randomization allows you to identify the ATE as the difference in outcome between treated and control, once you condition on the common effect, you also introduce selection bias. To see this, you can re-create this graph and delete the causal path from the new feature to customer satisfaction, which also closes the direct path to NPS. Then, you can check if NPS is still connected to the new features, once you condition on the response. You can see that it is, meaning that association flows between the two variables via a noncausal path, which is precisely what bias means:

```
In [13]: nps_model = nx.DiGraph([
             ("RND", "New Feature"),
```

```
#     ("New Feature", "Customer Satisfaction"),
    ("Customer Satisfaction", "NPS"),
    ("Customer Satisfaction", "Response"),
    ("New Feature", "Response"),
])


not(nx.d_separated(nps_model, {"NPS"}, {"New Featu
```

```
Out[13]: True
```

SEE ALSO

Causal identification under selection bias is very sneaky. This approach of deleting the causal path and checking if the treatment and outcome are still connected won't always work with selection bias. Unfortunately, at the time of this writing, I'm not aware of any Python libraries that deal with selection-biased graphs. But you can check DAGitty, which works on your browser and has algorithms for identification under selection bias.

To develop your intuition about this bias, let's get your godlike powers back and pretend you can see into the world of counterfactual outcomes. That is, you can see both the NPS customers would have under the control, NPS_0, and under the treatment, NPS_1, for all customers, even those who didn't answer the survey. Let's also simulate data in such a way that we know the true effect. Here, the new feature increases NPS by 0.4 (which is a high number for any business standards, but bear with me for the sake of the example). Let's also say that both the new feature and customer satisfaction increases the chance of responding to the NPS survey, just like we've shown in the previous graph. With the power to measure counterfactuals, this is what you would see if you aggregated the data by the treated and control groups:

	responded	nps_0	nps_1	nps
new_feature				
0	0.183715	–0.005047	0.395015	–0.0050
1	0.639342	–0.005239	0.401082	0.40108

First, notice that 63% of those with the new feature responded to the NPS survey, while only 18% of those in the control responded to it. Next, if you look at both treated and control rows, you'll see an increase of 0.4 by going from NPS_0 to NPS_1. This simply means that the effect of the new feature is 0.4 for both groups. Finally, notice that the difference in NPS between treated (new_feature=1) and control (new_feature=0) is about 0.4. Again, if you could see the NPS of those who did not respond to the NPS survey, you could just compare treated and control groups to get the true ATE.

Of course, in reality, you can't see the columns NPS_0 and NPS_1. You also also can't see the NPS column like this, because you only have NPS for those who responded to the survey (18% of the control rows and 63% of the treated rows):

	responded	nps_0	nps_1	nps
new_feature				
0	0.183715	NaN	NaN	NaN
1	0.639342	NaN	NaN	NaN

If you further break down the analysis by respondents, you get to see the NPS of those where $Response = 1$. But notice how the difference between treated and control in that group is no longer 0.4, but only about half of that (0.22). How can that be? This is all due to selection bias:

		nps_0	nps_1	nps
responded	**new_feature**			
0	**0**	NaN	NaN	NaN
	1	NaN	NaN	NaN
1	**0**	NaN	NaN	0.31407
	1	NaN	NaN	0.53610

Adding back the unobservable quantities, you can see what is going on (focus on the respondents group here):

		nps_0	nps_1	nps
responded	**new_feature**			
0	**0**	–0.076869	0.320616	–0.0768
	1	–0.234852	0.161725	0.16172
1	**0**	0.314073	0.725585	0.31407
	1	0.124287	0.536106	0.53610

Initially, treated and control groups were comparable, in terms of their baseline satisfaction Y_0. But once you condition on those who responded the survey, the treatment group has lower baseline satisfaction $E[Y_0|T=0,R=1] > E[Y_0|T=1,R=1]$. This means that a simple difference in averages between treated and control does not identify the ATE, once you condition on those who responded:

$$
\begin{aligned}
E[Y|T=1,R=1] - E[Y|T=1,R=1] &= \underbrace{E[Y_1 - Y_0|R=1]}_{ATE} \\
&+ \underbrace{E[Y_0|T=0,R=1] - E[Y_0|T=1,R=1]}_{\text{SelectionBias}}
\end{aligned}
$$

That bias term won't be zero if the outcome, customer satisfaction, affects the response rate. Since satisfied customers are more likely to answer the NPS survey, identification is impossible in this situation. If the treatment increases satisfaction, then the control group will contain more customers whose baseline satisfaction is higher than the treatment group. That's because the treated group will have those who were satisfied (high baseline satisfaction) plus those who had low baseline satisfaction, but due to the treatment, became more satisfied and answered the survey.

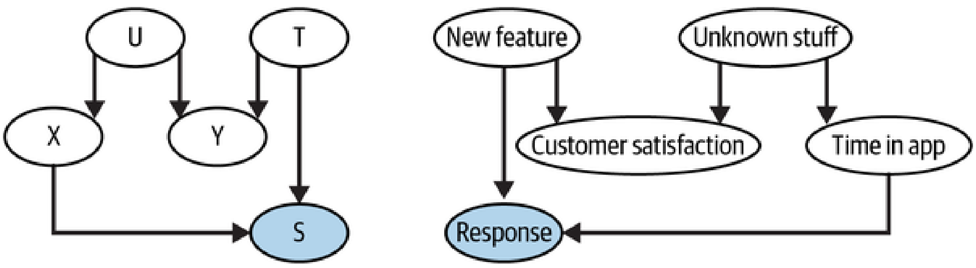
SEE ALSO

Selection bias is a much more complex topic than I can give it justice in this chapter. For instance, you can have selection bias simply by conditioning on an effect of the outcome, even if that effect isn't shared with the treatment. This situation is called a *virtual collider*. To learn more about it and much more, I strongly recommend you check out the paper "A Crash Course in Good and Bad Controls," by Carlos Cinelli et al. It goes through everything covered in this chapter and more. The paper is also written in clear language, making it easy to read.

Adjusting for Selection Bias

Unfortunately, correcting selection bias is not at all trivial. In the example we've been discussing, even with a randomized control trial, the ATE is not identifiable, simply because you can't close the noncausal flow of association between the new feature and customer satisfaction, once you condition on those who responded to the survey. To make some progress, you need to make further assumptions, and here is where the graphical model starts to shine. It allows you to be very explicit and transparent about those assumptions.

For instance, you need to assume that the outcome doesn't cause selection. In our example, this would mean that customer satisfaction doesn't cause customers to be more or less likely to answer the survey. Instead, you would have some other *observable* variable (or variables) that cause both selection and the outcome. For example, it could be that the only thing that causes customers to respond to the survey is the time they spend in the app and the new feature. In this case, the noncausal association between treatment and control flows through time in the app:

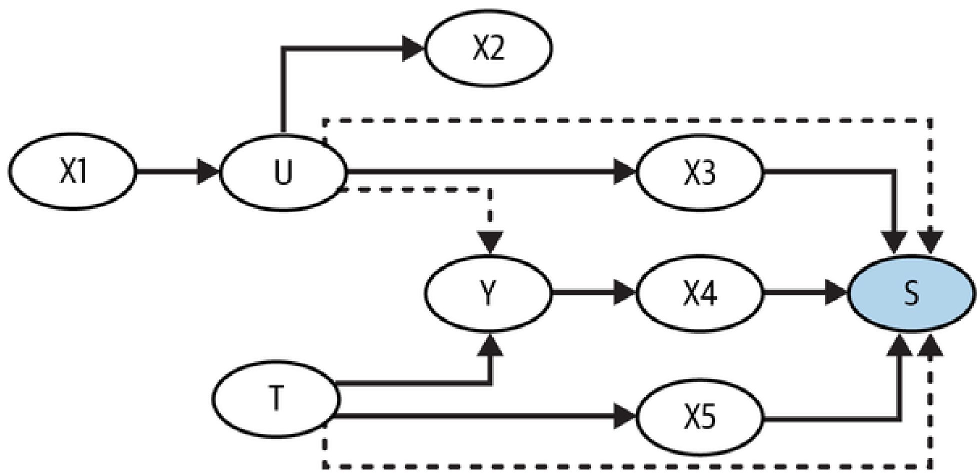


Only expert knowledge will be able to tell how strong of an assumption that is. But if it is correct, the effect of the new feature on satisfaction becomes identifiable once you control for time in the app.

Once again, you are applying the adjustment formula here. You are simply segmenting the data into groups defined by X so that treated and control groups become comparable within those segments. Then, you can just compute the weighted average of the in-group comparison between treated and control, using the size of each group as the weights. Only now, you are doing all of this while also conditioning on the selection variable:

$$ATE = \sum_x \{(E[Y|T=1, R=1, X] - E[Y|T=0, R=1, X])P(X|R=1)\}$$

Generally speaking, to adjust for selection bias, you have to adjust for whatever causes selection and you also have to assume that neither the outcome nor the treatment causes selection directly or shares a hidden common cause with selection. For instance, in the following graph, you have selection bias since conditioning on S opens a noncausal association path between T and Y:



You can close two of these paths by adjusting for measurable variables that explain selection, $X3$, $X4$, and $X5$. However, there are two paths you cannot close (shown in dashed lines): $Y \rightarrow S \leftarrow T$ and $T \rightarrow S \leftarrow U \rightarrow Y$. That's because the treatment causes selection directly and the outcome shares a hidden common cause with selection. You can mitigate the bias from this last path by further conditioning on $X2$ and $X1$, as they account for some variation in U, but that will not eliminate the bias completely.

This graph reflects a more plausible situation you will encounter when it comes to selection bias, like the response bias we've just used as an example. In these situations, the best you can do is to condition on variables that explain the selection. This will reduce the bias, but it won't eliminate it because, as you saw, 1) there are things that cause selection that you don't know or can't measure, and 2) the outcome or the treatment might cause selection directly.

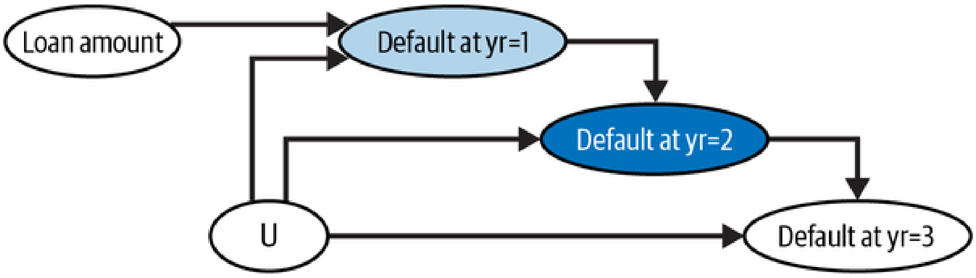
RACTICAL EXAMPLE

THE HIDDEN BIAS IN SURVIVAL ANALYSIS

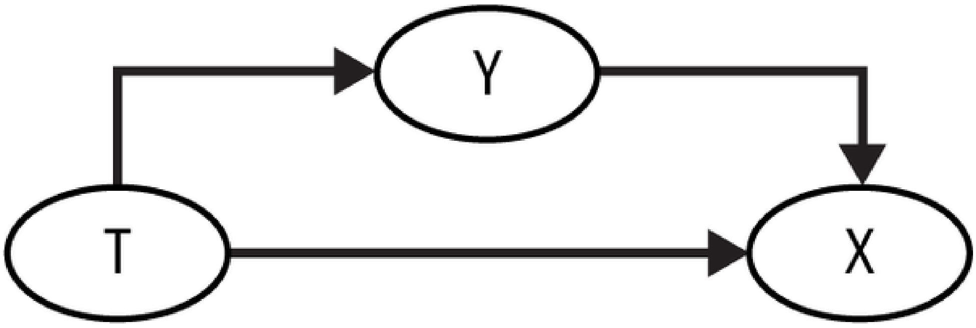
Survival analysis appears in many business applications that involve duration or time to an event. For instance, a bank is very interested in understanding how the size of a loan (loan amount) increases the chance of a customer defaulting on that loan. Consider a 3-year loan. The customer can default in the first, second, or third year, or they could not default at all. The goal of the bank is to know how loan amount impacts $P(Default \mid \mathrm{yr} = 1)$, $P(Default \mid \mathrm{yr} = 2)$, and $P(Default \mid \mathrm{yr} = 3)$. Here, for simplicity's sake, consider that the bank has randomized the loan amount. Notice how only customers who survived (did not default) in year 1 are observed in year 2 and only customers who survived years 1 and 2 are observed in year 3. This selection makes it so that only the effect of loan size in the first year is identifiable.

Intuitively, even if the loan amount was randomized, it only stays that way in the first year. After that, if the loan amount increases the chance of default, customers with lower risk of defaulting will be overrepresented in the region with high loan amounts. Their risk will have to be low enough to offset the increase caused by a bigger loan amount; otherwise, they would have defaulted at year 1. If the bias is too extreme, it can even look like bigger loans cause risk to decrease in a year after year 1, which doesn't make any sense.

A simple solution for this selection bias problem is to focus on cumulative outcome (survival), $Y|time > t$, rather than yearly outcomes (hazard rates), $Y|time = t$. For example, even though you can't identify the effect of loan amount on default at year 2, $P(Default \mid \mathrm{yr} = 2)$, you can easily identify the effect on default up to year 2, $P(Default \mid \mathrm{yr} \leq 2)$:



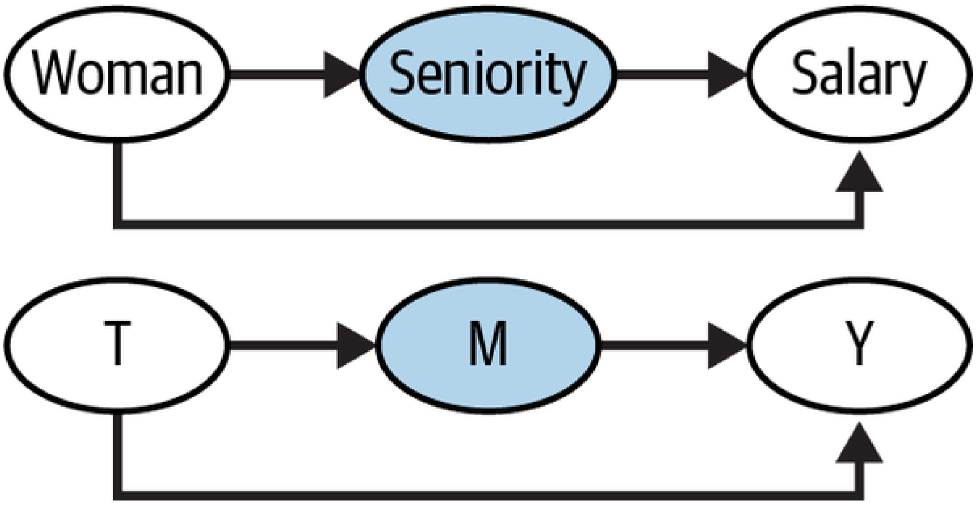
I also don't want to give you the false idea that just controlling for everything that causes selection is a good idea. In the following graph, conditioning on X opens a noncausal path, $Y \rightarrow X \leftarrow T$:



Conditioning on a Mediator

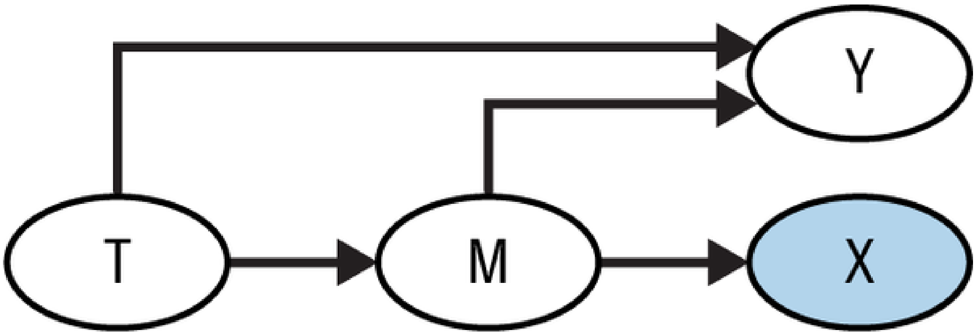
While the selection bias discussed so far is caused by unavoidable selection into a population (you were forced to condition on the respondent population), you can also cause selection bias inadvertently. For instance, let's suppose you are working in HR and you want to find out if there is gender discrimination; that is, if equally qualified men and women are paid differently. To do that analysis, you might consider controlling for seniority level; after all, you want to compare employees who are equally qualified, and seniority seems like a good proxy for that. In other words, you think that if men and women in the same position have different salaries, you will have evidence of a gender pay gap in your company.

The issue with this analysis is that the causal diagram probably looks something like this:



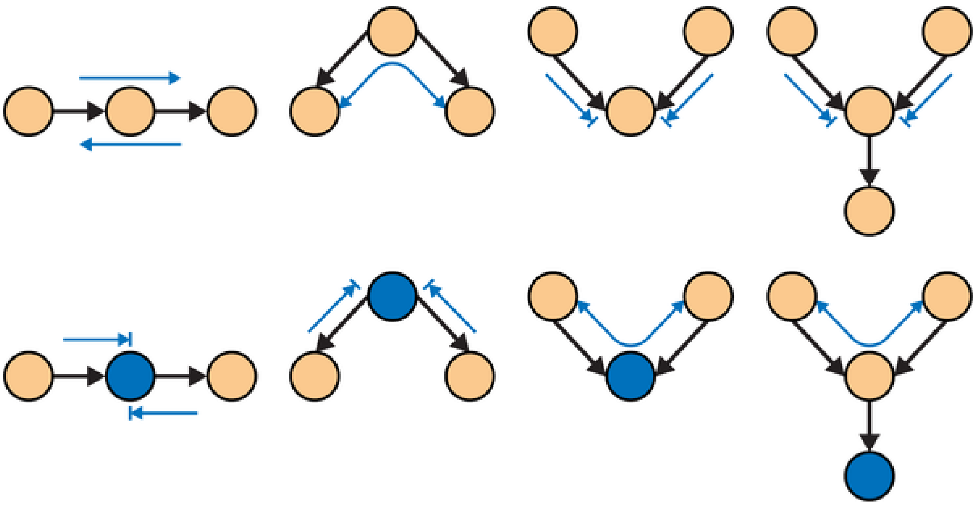
The seniority level is a *mediator* in the path between the treatment (woman) and salary. Intuitively, the difference in salary between women and men has a direct cause (the direct path, $woman \rightarrow salary$) and an indirect cause, which flows through the seniority (the indirect path $woman \rightarrow seniority \rightarrow salary$). What this graph tells you is that one way women can suffer from discrimination is by being less likely to be promoted to higher seniorities. The difference in salary between men and women is partly the difference in salary at the same seniority level, but also the difference in seniority. Simply put, the path $woman \rightarrow seniority \rightarrow salary$ is also a causal path between the treatment and the outcome, and you shouldn't close it in your analysis. If you compare salaries between men and women while controlling for seniority, you will only identify the direct discrimination, $woman \rightarrow salary$.

It is also worth mentioning that conditioning on descendants of the mediator node also induces bias. This sort of selection doesn't completely shut the causal path, but it partially blocks it:



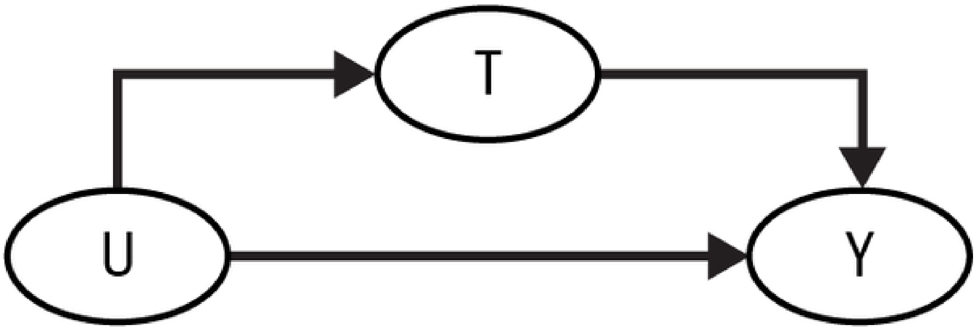
Key Ideas

In this chapter, you focused mostly on the identification part of causal inference. The goal was to learn how to use graphical models to be transparent about the assumptions you are making and to see what kind of association—causal or not—those assumptions entail. To do that, you had to learn how association flows in a graph. This cheat sheet is a good summary of those structures, so I recommend you keep it close by:



Then, you saw that identification amounts to isolating the causal flow of association from the noncausal ones in a graph. You could close noncausal paths of association by adjusting (conditioning) on some variables or even intervening on a graph, like in the case where you do a randomized experiment. Bayesian network software, like `networkx`, is particularly useful here, as it aids you when checking if two nodes are connected in a graph. For instance, to check for confounder bias, you can simply remove the causal path in a graph and check if the treatment and outcome nodes are still connected, even with that path removed. If they are, you have a backdoor path that needs to be closed.

Finally, you went through two very common structures of bias in causal problems. Confounding bias happens when the treatment and the outcome share a common cause. This common cause forms a fork structure, which creates a noncausal association flow between the treatment and the outcome:



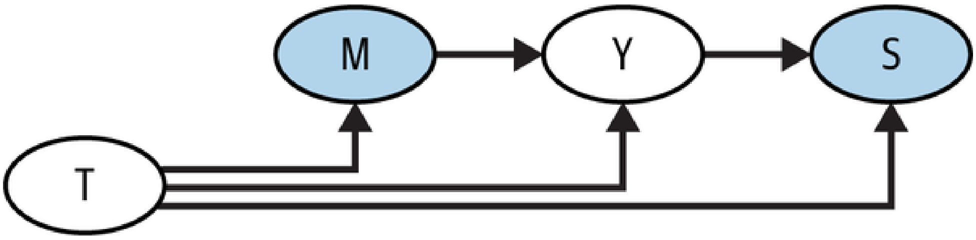
To fix confounding bias, you need to try to adjust for the common causes, directly or by the means of proxy variables. This motivated the idea of the adjustment formula:

$$ATE = \sum_x \{(E[Y|T=1, X=x] - E[Y|T=0, X=x])P(X=x)\},$$

and the conditional independence assumption, which states that, if treatment is as good as randomly assigned within groups of variables X, then you can identify causal quantities by conditioning on X.

Alternatively, if you can intervene on the treatment node, confounding becomes a lot easier to deal with. For instance, if you design a random experiment, you'll create a new graph where the arrows pointing to the treatment are all deleted, which effectively annihilates confounding bias.

You also learned about selection bias, which appears when you condition on a common effect (or descendant of a common effect) between the treatment and the outcome or when you condition on a mediator node (or a descendant of a mediator node). Selection bias is incredibly dangerous because it does not go away with experimentation. To make it even worse, it can be quite counterintuitive and hard to spot:



Again, it is worth mentioning that understanding a causal graph is like learning a new language. You'll learn most of it by seeing it again and again and by trying to use it.